Quick start for LaTeXing with the IEEEtran class for IEEE Computer Society Conferences

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***Abstract*—Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.**

I. Introduction

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.TODO!

The remainder of the paper starts with a presentation of related work (Section II). It is followed by a presentation of hints on LaTeX (Section III). The paper closes with a conclusion and an outlook on future work (Section IV).

II. Related Work

Winery [1] is a graphical modeling tool. The whole idea of TOSCA is explained by Binz et al. [2].

To search for related work, ScienceOS[1] can be used; it stores data in the EU and backs its answers with citations, which you still have to read and check yourself. Writing hints for student theses are given in the open-access book *Student Thesis Projects*[2]. More hints are given in *Writing for Computer Science*[3].

III. LaTeX Hints

This section contains hints on writing LaTeX. It focuses on minimal examples, which can be directly adapted to the content

A. Handling of paragraphs

One sentence per line. This rule is important for the usage of version control systems. A new line is generated with a blank line. As you would do in Word: New paragraphs are generated by pressing enter. In LaTeX, this does not lead to a new paragraph as LaTeX joins subsequent lines.

[1] https://www.scienceos.ai/
[2] https://vdf.ch/product/student-thesis-projects-en.html
[3] https://doi.org/10.1007/978-1-4471-6639-9

To start a new paragraph, press enter twice. This leads to an empty line. In word, there is the functionality to press shift and enter. This leads to a hard line break. The text starts at the beginning of a new line. In LaTeX, you can do that by using two backslashes (\\).
Running text seldom needs a hard line break.

Please do *not* use two backslashes for new paragraphs. For instance, this sentence belongs to the same paragraph, whereas the last one started a new one. Andrew Stacey gives a long motivation for that in the section "Line Wrapping" of his *Document Revision System*[4].

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
709 œĒúŰŏŐ
710 One sentence per line.
711 This rule is important for the usage of version
        ↪ control systems.
712 A new line is generated with a blank line.
713 As you would do in Word:
714 New paragraphs are generated by pressing enter.
715 In LaTeX, this does not lead to a new paragraph
        ↪ as LaTeX joins subsequent lines.
716 To start a new paragraph, press enter twice.
717 This leads to an empty line.
718 In word, there is the functionality to press
        ↪ shift and enter.
719 This leads to a hard line break.
720 The text starts at the beginning of a new line.
721 In LaTeX, you can do that by using two
        ↪ backslashes (\textbackslash\textbackslash)
        ↪ .
722 \\
723 Running text seldom needs a hard line break.
724
725 Please do \textit{not} use two backslashes for
        ↪ new paragraphs.
726 For instance, this sentence belongs to the same
        ↪ paragraph, whereas the last one started a
        ↪  new one.
727 Andrew Stacey gives a long motivation for that in
        ↪  the section \enquote{Line Wrapping} of
        ↪ his \href{https://loopspace.mathforge.org
        ↪ /HowDidIDoThat/TeX/VCS/#section.3}{\emph{
        ↪ Document Revision System}}\footnote{\url{
        ↪ https://loopspace.mathforge.org/
        ↪ HowDidIDoThat/TeX/VCS/}}.
```

B. *Notes separated from the text*

The package mindflow enables writing down notes and annotations in a way so that they are separated from the main text.

This is a small note.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
734 œĒúŰŏŐ
735 \begin{mindflow}
736 This is a small note.
737 \end{mindflow}
```

[4] https://loopspace.mathforge.org/HowDidIDoThat/TeX/VCS/

C. *Handling TODOs*

Marked text.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
742 œĒúŰŏŐ
743 \textmarker{Marked text.}
```

With `\textmarker`, only the text color is changed, since this also works well for individual words.

Marked text.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
748 œĒúŰŏŐ
749 \textcomment{Marked text.}{A comment on it.}
```

Manual marking for text that has been changed since the last version.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
752 œĒúŰŏŐ
753 \modified{Manual marking for text that has been
        ↪ changed since the last version.}
```

This is some text. Changed text.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
756 œĒúŰŏŐ
757 This is some text.
758 \change{FL1: text adjusted}{Changed text}.
```

Here is a comment.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
761 œĒúŰŏŐ
762 Here is a comment\sidecomment{Comment}.
```

TODO!

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
765 œĒúŰŏŐ
766 \todo{A lot of text still needs to be produced
        ↪ here}
```

D. Hyphenation

LaTeX automatically hyphenates words. When using microtype, there should be fewer hyphenations than in other settings. You might still need to tweak the hyphenation. Here are some hints:

In case you write “application-specific”, then the word will only be hyphenated at the dash. You can also insert `\allowbreak{}` into the word, as in application-specific, but this is much more effort.

You can now write words containing hyphens which are hyphenated at other places in the word. For that, write `"=` instead of `-`, as in application”=specific. This is enabled by an additional configuration of the babel package.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
776 œŒúŰőŐ
777 In case you write \enquote{application-specific},
        ↪ then the word will only be hyphenated at
        ↪ the dash.
778 You can also insert \verb1\allowbreak{}1 into the
        ↪ word, as in applica\allowbreak{}tion-
        ↪ specific, but this is much more effort.
779
780 You can now write words containing hyphens which
        ↪ are hyphenated at other places in the
        ↪ word.
781 For that, write \verb1"=1 instead of \verb1-1, as
        ↪ in application"=specific.
782 This is enabled by an additional configuration of
        ↪ the babel package.
```

E. Typesetting Units

Numbers can be written plain text (such as 100), by using the siunitx package as follows: 100 $\frac{\mathrm{km}}{\mathrm{h}}$, or by using plain LaTeX (and math mode): 100 $\frac{km}{h}$.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
787 œŒúŰőŐ
788 Numbers can be written plain text (such as 100),
        ↪ by using the \href{https://ctan.org/pkg/
        ↪ siunitx}{siunitx} package as follows:
789 \SI{100}{\km\per\hour},
790 or by using plain \LaTeX{} (and math mode):
791 $100 \frac{\mathit{km}}{h}$.
```

5 % of 10 kg

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
794 œŒúŰőŐ
795 \SI{5}{\percent} of \SI{10}{kg}
```

Numbers are automatically grouped: 123 456.

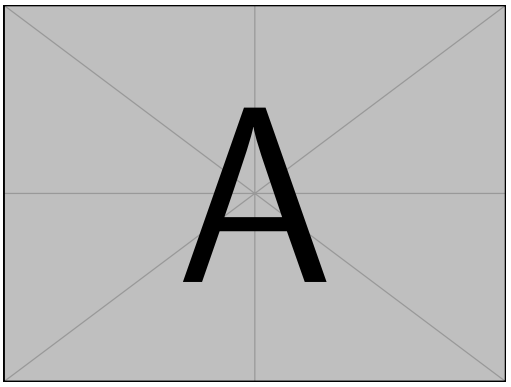


Figure 1. Example figure for zcref demo

Heading1	Heading2
One	Two
Three	Four

Figure 2. Example table for zcref demo

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
798 œŒúŰőŐ
799 Numbers are automatically grouped: \num{123456}.
```

F. Surrounding Text by Quotes

Please use the “enquote command” to quote something. Quoting with “quote” or “quote” also works.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
804 œŒúŰőŐ
805 Please use the \enquote{enquote command} to quote
        ↪ something.
806 Quoting with "`quote"' or ``quote'' also works.
```

G. Zref-clever examples

Demonstration of zref-clever: Zcref (short for zcref[S]) at the beginning of a sentence, zcref in all other cases.

Figure 1 shows a simple fact, although Figure 1 could also show something else.

Figure 2 shows a simple fact, although Figure 2 could also show something else.

Section III-G shows a simple fact, although Section III-G could also show something else.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
836 œŒúŰőŐ
837 \Zcref{fig:ex:cref} shows a simple fact, although
        ↪ \zcref{fig:ex:cref} could also show
        ↪ something else.
```

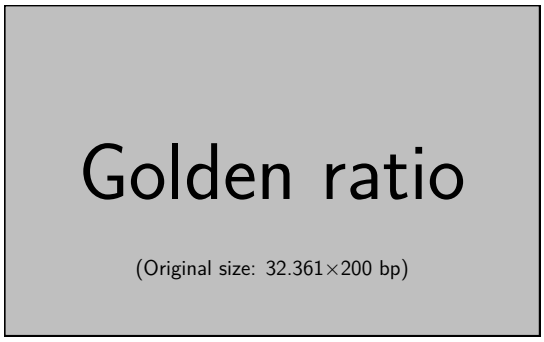


Figure 3. Simple Figure. Based on Scharrer [3].

```
\Zcref{tab:ex:cref} shows a simple fact, although \zcref{tab:ex:cref} could also show something else.

\Zcref{sec:ex:cref} shows a simple fact, although \zcref{sec:ex:cref} could also show something else.
```

H. Figures

Figure 3 shows something interesting.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\Zcref{fig:label} shows something interesting.

\begin{figure}
  \centering
  \includegraphics[width=.8\linewidth]{example-image-golden}
  \caption[Simple Figure]{
    Simple Figure.
    Based on \citet{mwe}.
  }
  \label{fig:label}
\end{figure}
```

Photos in JPEG or PNG format are included the same way, as shown in Figure 4.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{figure}
  \centering
  \includegraphics[width=.5\linewidth]{figures/ulm-minster.jpg}
  \caption{Ulm Minster. Photo: wkrauth, CC0, via Wikimedia Commons.}
  \label{fig:jpg}
\end{figure}
```

A figure spans multiple columns when using `\begin{figure*}`. See Figure 5 as an example.



Figure 4. Ulm Minster. Photo: wkrauth, CC0, via Wikimedia Commons.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{figure*}
  \centering
  % note that \textwidth is used instead of \linewidth
  % This ensures that the graphics width is 60% of the "page" (text block), and not just 60% of the current text column
  % See https://tex.stackexchange.com/a/17085/9075 for details
  \includegraphics[width=.6\textwidth]{example-image-16x9}
  \caption{16x9 Figure}
  \label{fig:16x9}
\end{figure*}
```

I. Sub Figures

An example of two sub figures is shown in Figure 6.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{figure*}[!t]
  \centering
  \subfloat[Case I]{\includegraphics[width=.4\linewidth]{example-image-a}%
    \label{fig:first_case}}
  \hfil
  \subfloat[Case II]{\includegraphics[width=.4\linewidth]{example-image-b}%
    \label{fig:second_case}}
  \caption{Example figure with two sub figures.}
  \label{fig:two_sub_figures}
\end{figure*}
```

Note that often IEEE papers with subfigures do not employ subfigure captions (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for the subfig package to generate the (a), (b), etc.,

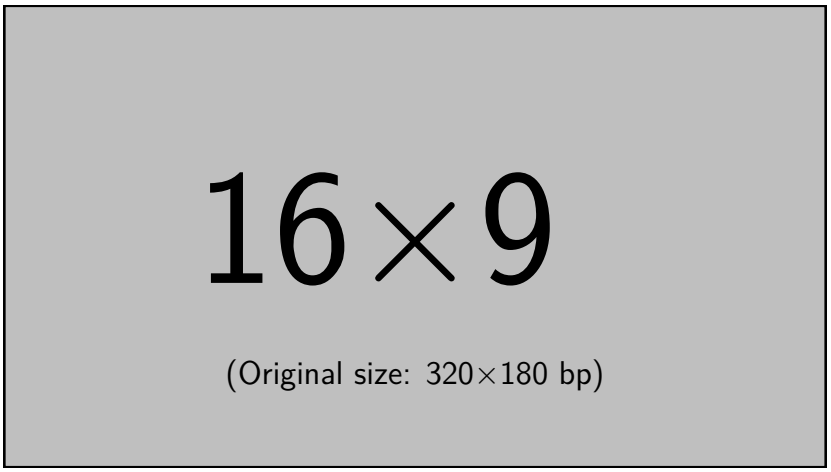
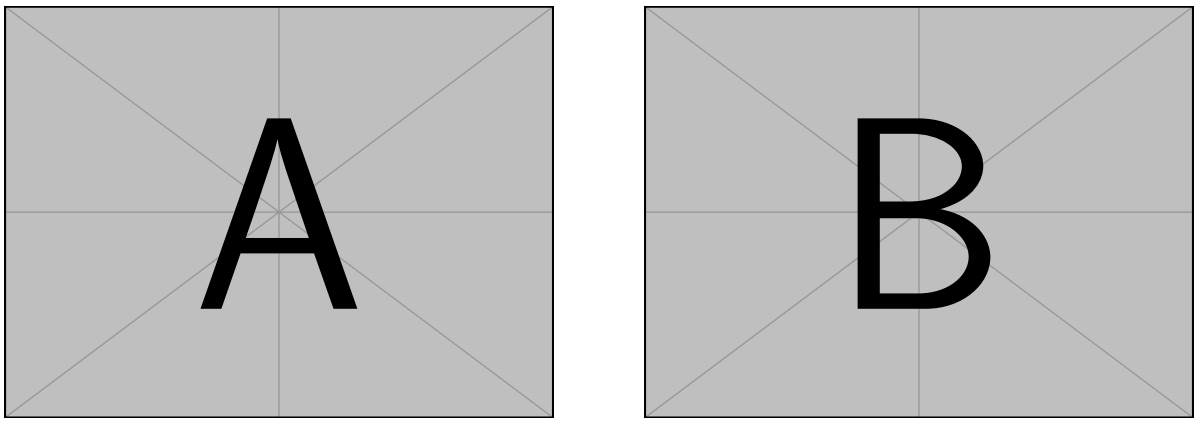


Figure 5. 16x9 Figure



(a) Case I (b) Case II

Figure 6. Example figure with two sub figures.

subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g., `\subfloat[]`. An example is shown in Figure 7.

Corresponding LaTeX code of `./paper-basic-listings.tex`

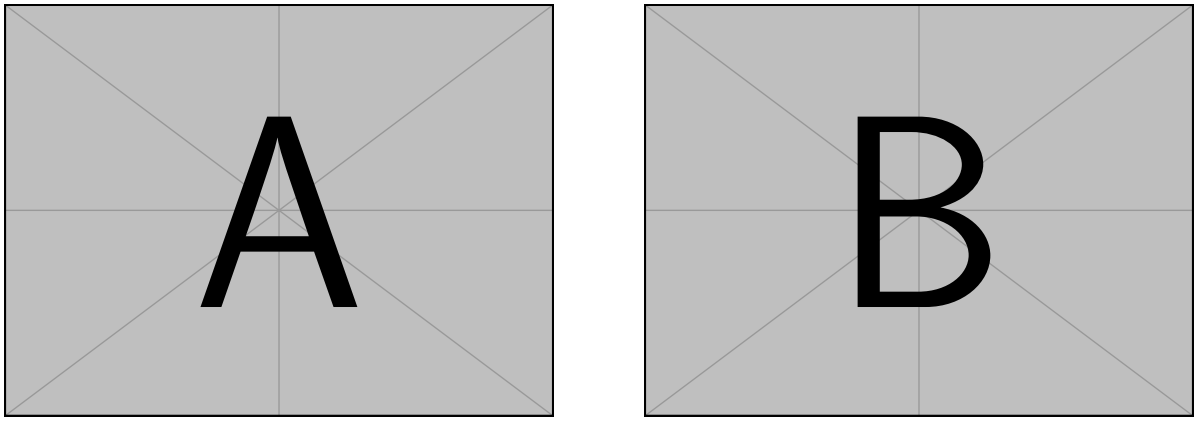
```
913 œŒűŰőŐ
914 \begin{figure*}[!t]
915   \centering
916   \subfloat[]{\includegraphics[width=.4\linewidth
          ↪ ]{example-image-a}%
917     \label{fig:first_case_ieee}}
918   \hfil
919   \subfloat[]{\includegraphics[width=.4\linewidth
          ↪ ]{example-image-b}%
920     \label{fig:second_case_ieee}}
921   \caption{Example figure with two sub figures.
          ↪ IEEE style. (a) The first case. (b) The
          ↪  second case.}
922   \label{fig:two_sub_figures_ieee}
923 \end{figure*}
```

J. *Figures with tikz*

The tikz package creates graphics programmatically. This package generates grids and other regular structures.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
931 œŒűŰőŐ
932 \begin{figure}[ht]
933   \centering
934   \begin{tikzpicture}
935     \draw(0,0) rectangle (4,4);
936     \foreach \x in {0.5,1,1.5,2,2.5,3,3.5}
937     \foreach \y in {0.5,1,1.5,2,2.5,3,3.5}
938     \draw(\x,\y) circle (1pt);
939   \end{tikzpicture}
940   \caption{A regular grid generated easily with
          ↪ two for loops.}\label{fig:tikz_example}
941 \end{figure}
```



(a) (b)

Figure 7. Example figure with two sub figures. IEEE style. (a) The first case. (b) The second case.

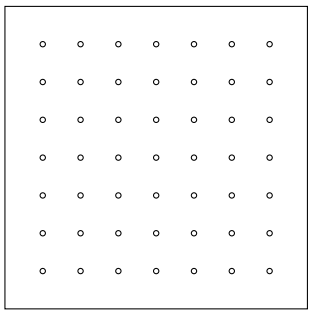


Figure 8. A regular grid generated easily with two for loops.

K. Plots with pgfplots

The pgfplots package plots functions directly in LaTeX, similar to MATLAB or gnuplot. Some visual examples are available in the package documentation[5].

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{figure}[ht]
  \centering
  \begin{tikzpicture}
    \begin{axis}[xlabel=$x$, ylabel=$\sin(x)$]
      \addplot {sin(deg(x))}; % Plot the sine
          ↪ function
    \end{axis}
  \end{tikzpicture}
  \caption{Plot of $\sin(x)$ directly inside the
      ↪ figure environment with pgfplots.}
  \label{fig:pgfplots_example}
\end{figure}
```

Coordinates can also be given explicitly instead of as a function:

[5] http://texdoc.net/pkg/pgfplots

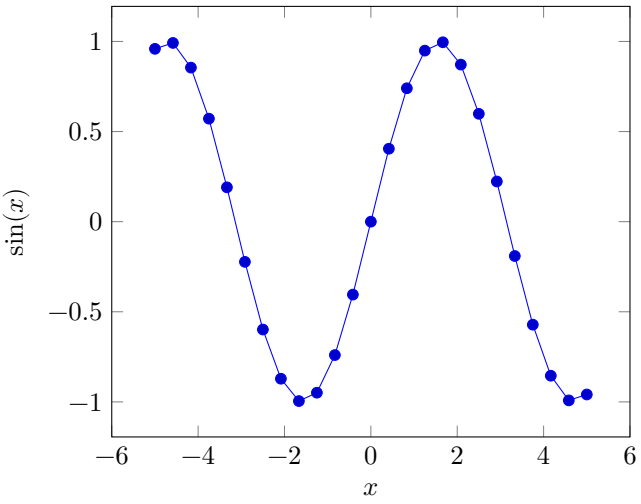


Figure 9. Plot of $\sin(x)$ directly inside the figure environment with pgfplots.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{figure}[ht]
  \centering
  \begin{tikzpicture}
    \begin{axis}[xlabel=$x$, ylabel=$y$]
      \addplot coordinates {(0,0) (1,1) (2,4) (3,
          ↪ 9)}; % Plot explicit coordinates
    \end{axis}
  \end{tikzpicture}
  \caption{Explicit coordinates plotted with
      ↪ pgfplots.}
  \label{fig:pgfplots_coords}
\end{figure}
```

L. Tables

Note that IEEE does not support `\begin{table}`, one has to use `\begin{figure}`.

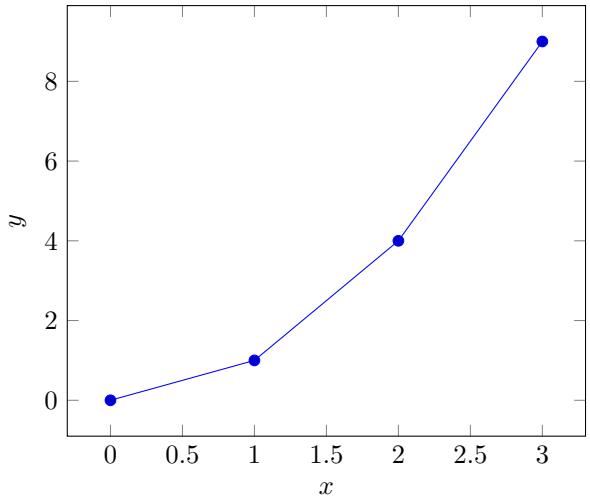


Figure 10. Explicit coordinates plotted with pgfplots.

Figure 11. Simple Table

Heading1	Heading2
One	Two
Three	Four

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĖúŰőŐ
\begin{figure}
  \caption{Simple Table}
  \label{tab:simple}
  \centering
  \begin{tabular}{ll}
    \toprule
    Heading1 & Heading2 \\
    \midrule
    One      & Two      \\
    Three    & Four     \\
    \bottomrule
  \end{tabular}
\end{figure}
```

Rows and cells can be colored with `\rowcolor` and `\cellcolor` from the colortbl package, which xcolor loads with its option `table`.

Figure 12. Table with colored cells

Test	Result
A	pass
B	fail

Figure 13. Table with diagonal line

Row \ Column	A	B
1	x	y

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĖúŰőŐ
\begin{figure}
  \caption{Table with colored cells}
  \label{tab:colored}
  \centering
  \begin{tabular}{ll}
    \toprule
    \rowcolor{gray!20}
    Test & Result                    \\
    \midrule
    A    & \cellcolor{green!20}pass \\
    B    & \cellcolor{red!20}fail  \\
    \bottomrule
  \end{tabular}
\end{figure}
```

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĖúŰőŐ
% Adapted from https://tex.stackexchange.com/a
    ↪ /468994/9075
\begin{figure}
  \caption{Table with diagonal line}
  \label{tab:diag}
  \begin{center}
    \begin{tabular}{|l|c|c|}
      \hline
      \diagbox{Row}{Column} & A & B \\
      \hline
      1                     & x & y \\
      \hline
    \end{tabular}
  \end{center}
\end{figure}
```

M. Source Code

Listing 1 shows source code written in XML. Line 2 contains a comment.

```
<listing name="example">
  <!-- comment -->
  <content>not interesting</content>
</listing>
```

Listing 1. Example XML Listing

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĖúŰőŐ
\Zcref{lst:XML} shows source code written in XML.
\Zcref{line:comment} contains a comment.

```

```
<listing name="example">
  Floating
</listing>
```

Listing 2. Example XML listing – placed as floating figure

```
{
  key: "value"
}
```

Listing 3. Example JSON listing – placed as floating figure

```
\begin{lstlisting}[
  language=XML,
  caption={Example XML Listing},
  label={lst:XML}]
<listing name="example">
  <!-- comment --> (* \label{line:comment} *)
  <content>not interesting</content>
</listing>
\end{lstlisting}
```

One can also add `float` as parameter to have the listing floating. Listing 2 shows the floating listing.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{lstlisting}[
  % one can adjust spacing here if required
  % aboveskip=2.5\baselineskip,
  % belowskip=-.8\baselineskip,
  float,
  language=XML,
  caption={Example XML listing -- placed as
      ↪ floating figure},
  label={lst:flXML}]
<listing name="example">
  Floating
</listing>
\end{lstlisting}
```

One can also typeset JSON as shown in Listing 3.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{lstlisting}[
  float,
  language=json,
  caption={Example JSON listing -- placed as
      ↪ floating figure},
  label={lst:json}]
{
  key: "value"
}
\end{lstlisting}
```

Java is also possible as shown in Listing 4.

```
public class Hello {
    public static void main (String[] args) {
        System.out.println("Hello World!");
    }
}
```

Listing 4. Example Java listing

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{lstlisting}[
  caption={Example Java listing},
  label=lst:java,
  language=Java,
  float]
public class Hello {
    public static void main (String[] args) {
        System.out.println("Hello World!");
    }
}
\end{lstlisting}
```

N. *Itemization*

One can list items as follows:

- Item One
- Item Two

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{itemize}
  \item Item One
  \item Item Two
\end{itemize}
```

With the package paralist, one can create itemizations with lesser spacing:

- Item One
- Item Two

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œŒúŰőŐ
\begin{compactitem}
  \item Item One
  \item Item Two
\end{compactitem}
```

One can enumerate items as follows:

1) Item One
2) Item Two

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĞúŰőŐ
\begin{enumerate}
  \item Item One
  \item Item Two
\end{enumerate}
```

With the package paralist, one can create enumerations with lesser spacing:

1) Item One
2) Item Two

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĞúŰőŐ
\begin{compactenum}
  \item Item One
  \item Item Two
\end{compactenum}
```

With paralist, one can even have all items typeset after each other and have them clean in the TeX document: 1) The first item, 2) the second item, and 3) the third item appear in one line.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĞúŰőŐ
\begin{inparaenum}
  \item The first item,
  \item the second item, and
  \item the third item appear in one line.
\end{inparaenum}
```

O. Writing in Markdown

If you are new to LaTeX, you can start by writing your text in Markdown using the markdown package. LaTeX then takes care of the typesetting. Markdown and LaTeX can be mixed: Table I is written in Markdown and referenced from LaTeX. A longer part, such as a whole section, can be placed in a separate file included by `\markdownInput{introduction.md}`. The markdown-latex-quickstart shows a complete document written in Markdown.

Text can be *emphasized* or **bold**, and `code` is set in a monospaced font. A citation looks like this [3], and here is a footnote.[6]

- first point
- second point
  - sub-point

1. firstly
2. secondly

[6]This is an inline footnote.

Right	Left	Centered
12	12	12
123	123	123

Table I
A TABLE WRITTEN IN MARKDOWN

Table I shows a table with caption. Its identifier `{#tbl:markdown}` is referenced by `<#tbl:markdown>`.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĞúŰőŐ
  \begin{markdown}
Text can be *emphasized* or **bold**, and `code`
     ↪ is set in a monospaced font.
A citation looks like this [@mwe], and here is a
     ↪ footnote.^[This is an inline footnote.]

- first point
- second point
  - sub-point

1) firstly
2) secondly

| Right | Left | Centered |
|------:|:-----|:--------:|
|    12 | 12   |    12    |
|   123 | 123  |   123    |

: A table written in Markdown {#tbl:markdown}

<#tbl:markdown> shows a table with caption.
Its identifier `{#tbl:markdown}` is referenced by
     ↪  `<#tbl:markdown>`.
\end{markdown}
```

P. Other Features

The words “workflow” and “dwarflike” can be copied from the PDF and pasted to a text file.

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĞúŰőŐ
The words \enquote{workflow} and \enquote{
     ↪ dwarflike} can be copied from the PDF and
     ↪  pasted to a text file.
```

The symbol for powerset is now correct: $\wp$ and not a Weierstrass p ($\wp$).

$\wp(1, 2, 3)$

Corresponding LaTeX code of `./paper-basic-listings.tex`

```
œĞúŰőŐ
The symbol for powerset is now correct: $\
     ↪ powerset$ and not a Weierstrass p ($\wp$)
     ↪ .

```

```
$\powerset({1,2,3})$
```

Brackets work as designed: <test> One can also input backticks in verbatim text: `test`.

Corresponding LaTeX code of ./paper-basic-listings.tex

```
œŒúŰőŐ
Brackets work as designed:
<test>
One can also input backticks in verbatim text: \verb|`test`|.
```

Acronyms can be typeset in running text with \initialism, which sets them as small caps that blend into the surrounding lowercase text. It is the lightweight alternative to the full acronym and glossary management (\gls, with an automatic list of abbreviations) provided by the thesis variants: \initialism only typesets the acronym and does not add it to any list. The macro and the ready-made acronyms (\OMG, \BPEL, \BPMN, \UML) are defined in commands.tex, where you can add your own.

The UML is maintained by the OMG; related standards include BPEL and BPMN. Any acronym can be set inline with REST or HTTP.

Corresponding LaTeX code of ./paper-basic-listings.tex

```
œŒúŰőŐ
The \UML is maintained by the \OMG; related standards include \BPEL and \BPMN.
Any acronym can be set inline with \initialism{REST} or \initialism{HTTP}.
```

IV. Conclusion and Outlook

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Acknowledgment

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which comes right before the reference section in your document [4].

References

[1] O. Kopp *et al.*, "Winery – A Modeling Tool for TOSCA-based Cloud Applications," in *Proceedings of 11th International Conference on Service-Oriented Computing (ICSOC'13)*, ser. LNCS, vol. 8274. Springer Berlin Heidelberg, 2013, pp. 700–704.

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[3] M. Scharrer, *The mwe Package*, 2017. [Online]. Available: http://texdoc.net/mwe

[4] B. Veytsman, "Latex class for the association for computing machinery – acknowledgement information," Aug. 2021. [Online]. Available: https://github.com/borisveytsman/acmart/blob/1704c8bf7eee92a1515ff755f5118b6a22bb1f8e/samples/samples.dtx#L709

All links were last followed on October 5, 2020.